
Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

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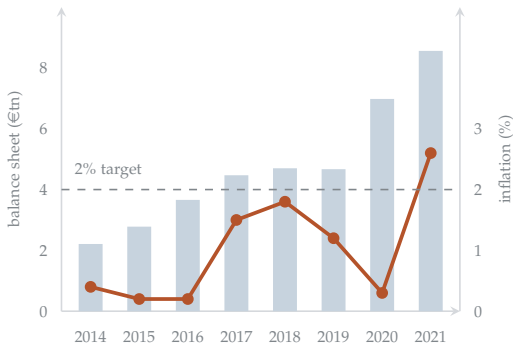
This talk follows one question from stimulus vouchers to the design of the digital euro

Should the digital euro bear interest, and if so, in what form?

- THE PROBLEM
Monetary policy stalls when new money is hoarded and interest rates cannot fall below the cash floor.
- A TEMPTING FIX
Vouchers with an expiry date get money spent, and a programmable digital euro could carry one.
- A BETTER FIX
An expiry date ends in a fire sale and splits money into vintages; continuous decay is simply a negative rate.
- THE TWIST
The draft regulation fixes the rate at zero, which makes the digital euro costless cash and raises the lower bound.
- THE DESIGN
Combining price and quantity suggests a free tier for payments, a Pigouvian rate above it, and a cap for crises.

When the central bank creates money that nobody spends, monetary policy loses its grip

Eurosystem balance sheet and inflation



Between 2014 and 2021 the Eurosystem almost quadrupled its balance sheet, from €2.2 trillion to €8.6 trillion.

Inflation nevertheless stayed below the 2% target in every year from 2014 to 2020, and was close to zero in 2014–16 and 2020.

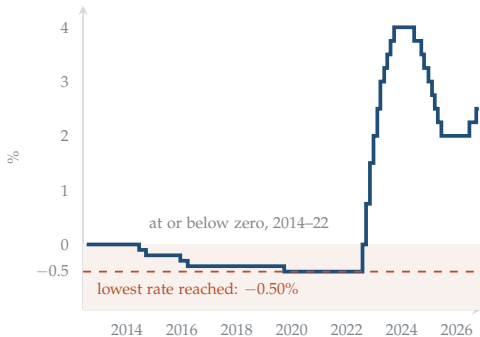
In the quantity equation $MV = PY$, the extra money M was offset by a fall in velocity V : households and firms held the new money instead of spending it.

This is the first blockage: money is created, but it is not spent.

Sources: ECB annual consolidated balance sheet of the Eurosystem (end of year); Eurostat HICP, annual average.

Interest rates could not be cut far below zero, because cash always pays zero

ECB deposit facility rate



Source: ECB key interest rates.

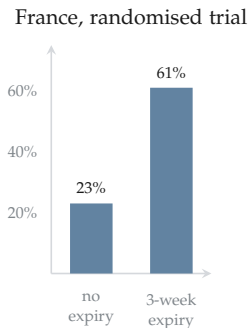
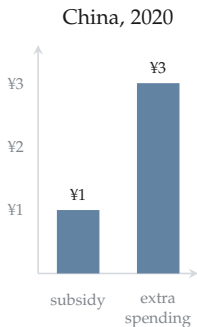
The ECB cut its deposit facility rate below zero in June 2014 and kept it at or below zero until July 2022, but it never went lower than -0.50% .

The reason is the outside option of cash. Banknotes pay zero, so if banks charged savers more than the cost of storing and insuring cash, savers would withdraw their money.

The lower bound is therefore not exactly zero but minus the cost of holding cash: $i \geq -c$.

This is the second blockage: interest rates cannot fall far enough to make people spend.

Vouchers with an expiry date get money spent, which is exactly what a liquidity trap needs



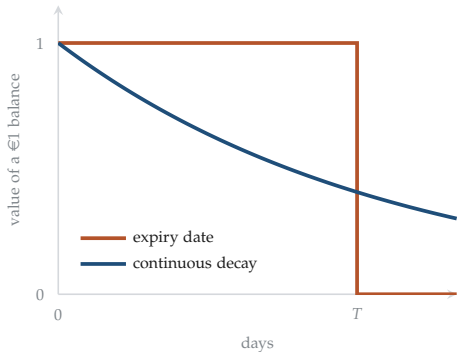
In 2020 Chinese cities distributed digital coupons through mobile payment platforms that had to be used within about a week. Each yuan of subsidy generated roughly three yuan of additional spending.

In a randomised experiment in France, households received prepaid cards. When the balance expired after three weeks, 61% was spent within a month, against 23% without an expiry date.

A deadline turns money that would have been saved into spending today.

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024), one-month MPC.

A programmable digital euro could build this pressure into money itself, in one of two ways



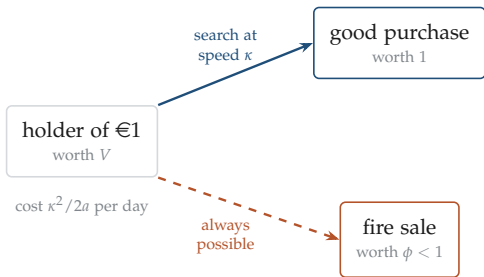
A central bank digital currency is programmable, so the central bank could charge holders for keeping money idle.

With an **expiry date**, nothing happens until day T , and then the whole balance is lost.

With **continuous decay**, also called demurrage, the balance shrinks by a small fraction every day, like Silvio Gesell's stamped money of the 1930s.

Both designs charge for holding money; they differ only in *when* the charge falls. **Which one should a central bank choose?**

To compare the two designs, consider a holder who must search for good uses of money



Each euro is worth V to its holder. Finding a good use for it takes effort: searching at speed κ costs $\kappa^2/2a$ per day, and κ is the velocity of money. A bad use, such as buying something of little value or selling the token at a discount, is always available and worth $\phi < 1$.

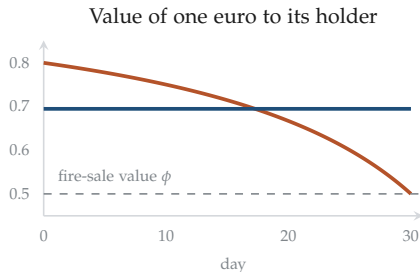
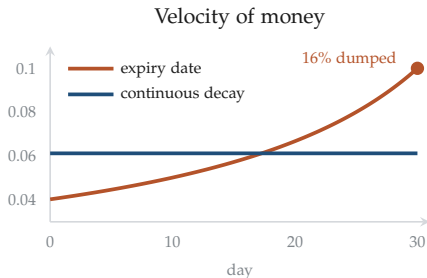
Decay at rate λ : $(r + \lambda) V = \frac{a}{2}(1 - V)^2$

Expiry at T : $\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2$, $V = \phi$ at T

In both cases the holder spends at speed $\kappa^* = a(1 - V)$.
The less money is worth holding, the faster it moves.

RESULT 1

An expiry date pushes out more spending, but it ends in a fire sale and splits money into vintages

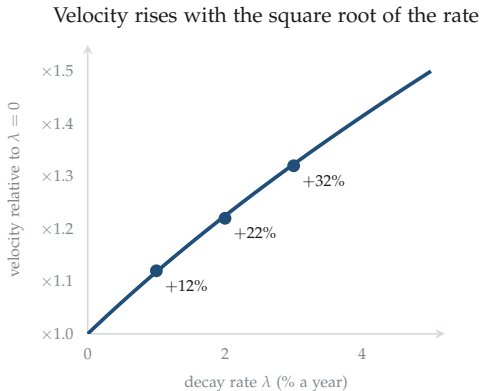


Both designs are calibrated to the same average velocity over 30 days. The expiry date gets more useful spending done, **84%** of issuance against **73%**, because it forces the whole balance out. But velocity accelerates into the deadline, and **16%** of issuance is still unspent on the last day and is dumped on goods worth half its value. Worse, an expiring euro is worth **0.80** when new and **0.50** just before it expires: one euro is no longer equal to another.

Parameters: $a = 0.2$, $\phi = 0.5$, $r = 4\%$ a year. Results at other horizons and parameters in the backup.

RESULT 2

Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate



Decay avoids every problem of the expiry date. There is no deadline and no fire sale, every euro has the same value whatever its age, and it can apply to all money rather than to a separate voucher.

A decay rate λ is simply an interest rate of $-\lambda$. Velocity follows $\kappa^* \approx \sqrt{2a(r + \lambda)}$, the same square-root rule as in the Baumol–Tobin model of money demand: a decay of 1% a year raises velocity by about 12%.

Decay addresses both blockages: money moves faster, and a negative rate finally reaches retail balances. An expiry date belongs to a voucher; decay can belong to money.

Yet the draft regulation fixes the digital euro's interest rate at exactly zero

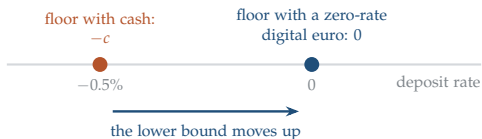
"The digital euro shall not bear interest."

European Commission, proposal for a regulation on the digital euro, Article 16(8), June 2023



The rate is not positive and not negative, in any circumstances. Holdings are also capped per person, with a range of €1,500–3,000 under discussion. The final text is being negotiated now, so the design question is still open.

A zero-rate digital euro is cash without storage costs, so it pushes the lower bound back up to zero



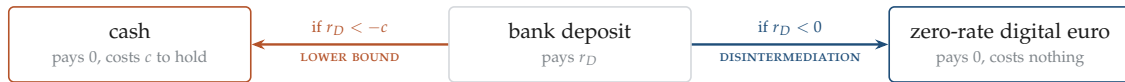
A bank cannot pay depositors less than their best alternative in central-bank money, net of the cost of holding it:

$$r_D \geq \max_j \{i_j - c_j\}.$$

Cash pays zero but costs something to store and insure, so the floor lies slightly below zero. This is why the ECB could reach -0.50% .

A zero-rate digital euro pays the same zero but costs nothing to hold. Up to the holding limit, it lifts the floor for deposit rates back to exactly zero, and takes away the little room below zero that the ECB had.

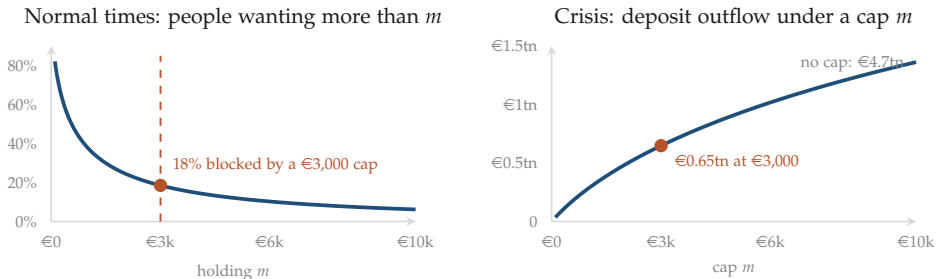
The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits



The lower bound and bank disintermediation are usually discussed as separate issues. They are the same mechanism: whenever central-bank money offers a better net return than a bank deposit, depositors move.

A zero-rate digital euro therefore puts the most pressure on bank deposits precisely when rates are negative, which is when banks' margins are already squeezed. **Fixing its rate at zero ties the digital euro's financial-stability risk to the state of monetary policy.**

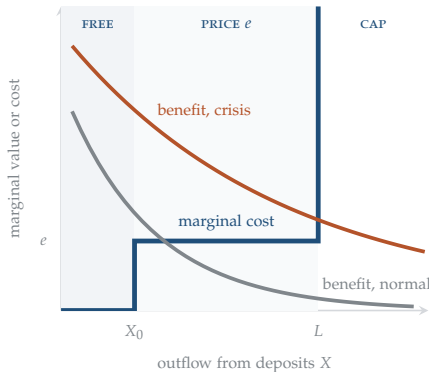
A hard holding limit stops a run in a crisis, but it also blocks ordinary users in normal times



In normal times people hold the digital euro to make payments. In the ECB's survey, 18% would like to hold more than €3,000; a hard cap simply blocks them, while a price would let them hold more if they pay for it. In a crisis people run for safety, and a small interest-rate difference cannot stop them. **Price is enough in normal times; only quantity works in a crisis.**

Survey distribution: Lambert et al. (ECB WP 2980, 2024). Outflows: own calibration to ECB technical data on holding limits (Oct 2025), which reports €699bn at €3,000.

When costs have a cliff and benefits do not, price and quantity together beat either one alone



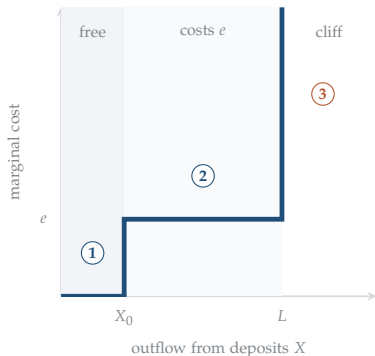
Weitzman (1974) showed that under uncertainty a price works better when the benefit curve is steeper than the cost curve, and a quantity works better when the cost curve is steeper:

$$\Delta_{\text{price-quantity}} = \frac{\sigma^2 (b - c)}{2 b^2}.$$

The marginal cost of deposit outflows has three segments: zero while banks can absorb them, a constant e while banks replace deposits with dearer funding, and vertical once their liquidity buffers run out.

In normal times demand crosses the flat part, where a price works best. In a crisis it shifts out and hits the cliff, where only a quantity works. Both happen on the same curve, which points to a tiered price with a cap (Roberts and Spence 1976).

Each segment of the cost curve becomes one rule: a free tier, a Pigouvian rate, and a cap



1. A free tier M_1 . Outflows up to X_0 cost banks nothing, so balances held for payments can be free. The tier is set so that payment balances just fill this free capacity:

$N \mathbb{E}[\min(t, M_1)] = X_0$, where t is what a person wants to hold.

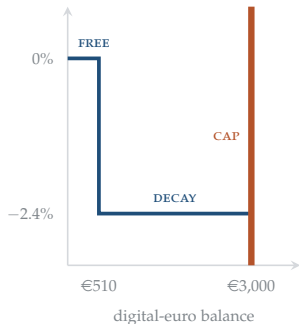
2. A Pigouvian rate $-e$ above the tier. Each further euro moved out of a bank costs the bank e to replace. Charging holders exactly this cost makes them weigh it, as a Pigouvian tax does: $i = -e$.

3. A cap $\bar{M}$. Beyond the liquidity buffer L the cost has no bound, so no price is high enough. Only a quantity limit keeps the crisis outflow within the buffer:

$N \mathbb{E}[\min(d, \bar{M})] = L$, where d is a person's sight deposit, all of which may run.

In an illustrative calibration, the rules imply a free tier of about €500, a rate of about -2.4% and a cap near €3,000

The resulting rate schedule



Rule	What the data say	Result
1. Free tier	The ECB expects payment digitalisation to bring €127bn of new deposits by 2034, so $X_0 = €127\text{bn}$. Desired holdings t : median €560, mean €3,220.	€510
2. Rate	A bank replaces a lost deposit at the refinancing rate (2.65%) instead of the deposit rate (0.26%), so e is 2.4 points.	-2.4%
3. Cap	Sight deposits d : median €2,000, mean €13,100. L is the crisis outflow the ECB judged manageable in October 2025.	€3,000

The cap is anchored to the ECB's own analysis. The free tier and the rate follow from the model and depend on its assumptions; **they show orders of magnitude, not a recommendation**. Details and sensitivity are in the appendix.

Data: Lambert et al. (ECB WP 2980, 2024), survey distributions; ECB technical data on holding limits (Oct 2025); ECB key rates (Sep 2026); MIR statistics (Apr 2026).

Population $N = 360\text{m}$, fitted to the ECB's outflow estimates.

CONCLUSION

If money is to carry a holding cost, continuous decay with a free tier and a cap is its natural form

1. Deadlines get money spent, but in the model an expiry date ends in a fire sale and makes one euro worth more than another. It suits a voucher better than money.
2. Continuous decay keeps every euro equal and is equivalent to a negative interest rate, so it is the natural form for a holding cost on money.
3. A digital euro fixed at zero behaves like cash without storage costs. It tends to raise the lower bound, and its pressure on bank deposits is greatest when rates are negative.
4. Weitzman's logic suggests combining price and quantity: a free tier for payments, a rate above it, and a cap for crises.

OPEN QUESTIONS

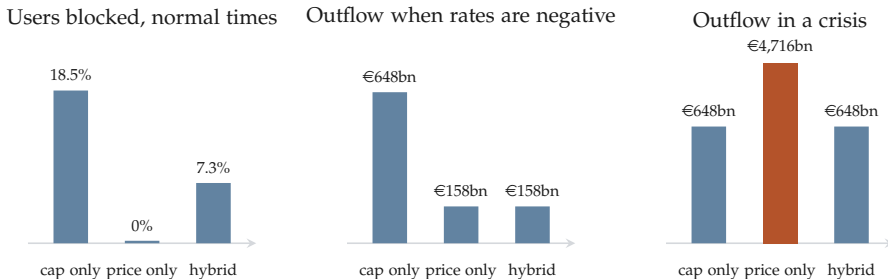
General-equilibrium effects on prices and output; the social rather than private cost of deposit outflows; behavioural responses to a shrinking balance; and the legal and political feasibility of a negative rate on public money.

Appendix

Related literature and how this talk relates to it

Strand	Main references	Use in this talk
Negative interest on money	Gesell (1916); Fisher (1933); Keynes (1936); Goodfriend (2000); Buiter & Panigirtzoglou (2003)	decay as a carry tax
Interest-bearing CBDC and the lower bound	Bordo & Levin (2017); Agarwal & Kimball (2019); Assenmacher & Krogstrup (2018)	decay removes the cash floor only with cash reform
Zero-rate CBDC and the lower bound	ECB (2020); Pirgmann & Wawrosz (2024)	slides 11–12
Tiered remuneration	Bindseil (2020); Bindseil & Panetta (2020)	free tier and rate above it
Holding limits	Meller & Soons (2023); ECB (2025); Lambert et al. (2024)	data for the calibration
Expiring CBDC	Kahn, van Oordt & Zhu (2026), for loss recovery	different purpose
Voucher evidence	Xing et al. (2023); Ding et al. (2024); Boehm, Fize & Jaravel (2024); Hsieh et al. (2010)	slide 5
Prices versus quantities	Weitzman (1974); Roberts & Spence (1976)	slide 14

Comparing designs: the hybrid performs well in normal times, at negative rates and in a crisis



A cap alone blocks almost one user in five and still lets €648bn leave banks whenever deposit rates turn negative, because every capped euro is a zero-rate euro. A price alone handles normal times and negative rates, but in a crisis it would let €4.7 trillion run. **The hybrid keeps the cap's protection in a crisis, the price's protection at negative rates, and blocks fewer users**, because the decay above the tier shrinks demand for large balances.

The price-only crisis bar is truncated. Model calibrated to ECB survey and holding-limit data; see backup.

Expiry versus decay: derivation and results

Holder's problem. Per unit of balance, with decay λ :

$$(r + \lambda)V = \max_{\kappa} \left\{ \kappa(1 - V) - \frac{\kappa^2}{2a} \right\} \Rightarrow \kappa^* = a(1 - V), \quad (r + \lambda)V = \frac{a}{2}(1 - V)^2.$$

Square-root rule. With $x = 1 - V$ small, $(r + \lambda)(1 - x) = \frac{a}{2}x^2$, so $x \approx \sqrt{2(r + \lambda)/a}$ and $\kappa^* \approx \sqrt{2a(r + \lambda)}$.

Expiry. With $\tau = T - t$ left: $dV/d\tau = -rV + \frac{a}{2}(1 - V)^2$, $V(0) = \phi$. If $\frac{a}{2}(1 - \phi)^2 > r\phi$, $V > \phi$ before T , so the fire sale happens only at T . Velocity rises to $a(1 - \phi)$: finite, but the unspent balance is sold at once. Decay λ is set to match the average expiry velocity over $[0, T]$.

T	useful (expiry)	dumped	value range	λ p.a.	useful (decay)	decayed
7 d	45%	55%	0.50–0.63	1172%	41%	15%
30 d	84%	16%	0.50–0.80	489%	73%	16%
90 d	97%	3%	0.50–0.91	161%	88%	10%

Matching a voucher needs decay of hundreds of percent a year; policy-sized decay of 1–3% raises velocity by 12–32%. A digital euro can carry a rate, not a voucher.

Prices versus quantities: derivation

Benefit $MB(q) = b_0 + \theta - bq$, cost $MC(q) = c_0 + cq$, shock θ with variance σ^2 ; optimum $q^*(\theta) = (b_0 + \theta - c_0)/(b + c)$.

- Quantity fixed at $\mathbb{E}q^*$: expected loss $\frac{\sigma^2}{2(b+c)}$
- Price fixed: q moves by θ/b , expected loss $\frac{\sigma^2 c^2}{2b^2(b+c)}$
- Difference: $\Delta = \frac{\sigma^2(b-c)}{2b^2}$. Prices win if and only if $b > c$.

Three-segment cost. $MC(X) = 0$ for $X \leq X_0$, e for $X_0 < X \leq L$, ∞ beyond L . The planner's optimum is decentralised by a zero rate up to M_1 , the rate $-e$ above it, and a cap at $\bar{M}$:

$$N \mathbb{E}[\min(t, M_1)] = X_0, \quad i_2 = -e, \quad N \mathbb{E}[\min(d, \bar{M})] = \theta D.$$

Under the rate $-e$, Baumol–Tobin demand for tier-2 balances shrinks by $K = \sqrt{r_D/(r_D + e)} = 0.31$.

Calibration: data and fit

Sight deposits per person	median €2,000, mean €13,100	ECB WP 2980
Desired digital-euro holdings	median €560, mean €3,220	ECB WP 2980
Crisis outflow	€156bn at €500; €699bn at €3,000	ECB, Oct 2025
Digitalisation inflow by 2034	€127bn	ECB, Oct 2025
MRO; household overnight rate	2.65%; 0.26%	ECB, Sep / Apr 2026
Aggregate LCR	156.7%	ECB, Q3 2025

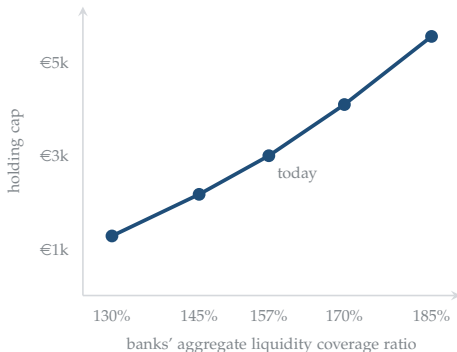
Fit. Lognormals from the survey; population fitted to the two outflows: $N = 360\text{m}$ (euro area 2026: 359.6m). Model: €156bn at €500, €648bn at €3,000; mean holding under a €3,000 cap €1,086 (survey: €1,100); 18% want more than €3,000 (survey: 20%). No cap: €4.7tn (Meller & Soons 2023: almost €5tn).

Sensitivity. X_0 of €100bn / 200bn / 300bn gives M_1 of €370 / 960 / 1,840. Tolerance θ of 5% / 7.6% / 10% gives $\bar{M}$ of €1,690 / 3,000 / 4,520.

Caveat. e is the private cost of replacing deposits; with central-bank pass-through funding the social cost may be lower (Brunnermeier & Niepelt 2019).

Extension: the cap could move with banks' liquidity

A countercyclical cap



The cap protects banks' liquidity, so it should depend on how much liquidity banks hold.

If the tolerated crisis outflow is proportional to banks' liquidity surplus above the 100% minimum, the cap becomes a function of the aggregate liquidity coverage ratio. At today's 157% it is €3,000. It would fall to about €1,300 at 130% and rise to about €5,600 at 185%.

The rule should be published and mechanical, so that a lower cap is anticipated rather than read as a signal of panic.

Aggregate LCR of significant institutions: 156.7% (ECB, Q3 2025). Scaling rule is illustrative.

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